

Pràctica Símplex: Resolució de Problemes

Membres del grup

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Assignatura / Curs

Optimització, Grau en Intel·ligència Artificial

Data

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1. Membres del grup

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2. Nombre de conjunts de dades usats

S'han utilitzat **dos conjunts de dades**: el conjunt **34** i el conjunt **37**, resolent un total de 8 problemes d'optimització.

3. Descripció resumida de la implementació

3. Descripció resumida de la implementació

Hem implementat la solució en Python a través de la classe `SimplexSolver`, aplicant el **mètode del símplex primal estructurat en dues fases** i utilitzant la **regla de Bland** en totes les decisions per evitar la degeneració.

La seqüència d'execució del nostre codi és la següent:

1. **Preprocessament:** Forcem els termes independents a ser no negatius ($b_i \geq 0$) canviant el signe de tota la fila corresponent de la matriu A si és necessari. Ademés, comprovem si la matriu conté una base trivial (identitat), cosa que ens permet passar a la fase II directament.
2. **Fase I (Solució Bàsica Factible Inicial):** Per assegurar l'arrencada independentment de les restriccions del problema, estenem la matriu afegint variables artificials (una matriu identitat) que componen la base inicial. Executem el nucli del símplex amb una funció objectiu que minimitza la suma d'aquestes variables artificials. Si el valor òptim d'aquesta fase és major que 0 (considerant una tolerància de 10^{-8}), declarem el problema com a infactible. En cas de degeneració (variables artificials encara a la base amb valor zero), intentem pivotar-les fora abans de passar a la fase II.
3. **Fase II (Optimització real):** Un cop garantida la factibilitat, eliminem les variables artificials no bàsiques i restablim el vector de costos original c . A partir d'aquí, continuem amb el símplex fins trobar una solució òptima o detectar que el problema és no acotat.
4. **Nucli iteratiu:** En cada iteració, calculem la solució bàsica $x_B = B^{-1} b$, els multiplicadors duals $w = c_B^T B^{-1}$ i la direcció $d_B = -B^{-1} A_q$.
5. **Pivoteig (Regla de Bland):** Per entrar a la base calculem els costos reduïts (r_j) manualment; d'entre els que són negatius (més enllà del llindar de tolerància), seleccionem el candidat amb l'índex j més petit. Per sortir de la base apliquem el test del quocient $\min \{x_B/d\}$ considerant $d > 0$, escollint novament l'índex més baix en cas d'empat per complir amb la Regla de Bland.

4. Solució obtinguda

A continuació es detalla el registre de l'execució pas a pas de cada iteració i les solucions òptimes per als 8 problemes dels conjunts indicats.

Conjunt 34

Problema 1

```
[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 7, q = 1, B(p) = 27, theta*= 0.659,
z = 2815.000
[jh_simplexP] Iteracio 2 : iout = 6, q = 2, B(p) = 26, theta*= 0.845,
z = 2707.000
[jh_simplexP] Iteracio 3 : iout = 3, q = 3, B(p) = 23, theta*= 1.218,
z = 2443.242
[jh_simplexP] Iteracio 4 : iout = 9, q = 4, B(p) = 29, theta*= 1.161,
z = 1991.135
[jh_simplexP] Iteracio 5 : iout = 6, q = 5, B(p) = 2, theta*= 1.134,
z = 1624.557
[jh_simplexP] Iteracio 6 : iout = 3, q = 6, B(p) = 3, theta*= 1.349,
z = 1468.862
[jh_simplexP] Iteracio 7 : iout = 8, q = 7, B(p) = 28, theta*= 0.383,
z = 1415.671
[jh_simplexP] Iteracio 8 : iout = 10, q = 3, B(p) = 30, theta*= 2.740,
z = 1222.967
[jh_simplexP] Iteracio 9 : iout = 5, q = 8, B(p) = 25, theta*= 1.514,
z = 957.663
[jh_simplexP] Iteracio 10 : iout = 3, q = 9, B(p) = 6, theta*= 2.064,
z = 735.863
[jh_simplexP] Iteracio 11 : iout = 4, q = 10, B(p) = 24, theta*= 1.571,
z = 682.006
[jh_simplexP] Iteracio 12 : iout = 1, q = 2, B(p) = 21, theta*= 0.057,
z = 474.694
[jh_simplexP] Iteracio 13 : iout = 1, q = 6, B(p) = 2, theta*= 0.150,
z = 463.824
[jh_simplexP] Iteracio 14 : iout = 8, q = 12, B(p) = 7, theta*= 0.717,
z = 463.043
[jh_simplexP] Iteracio 15 : iout = 3, q = 11, B(p) = 9, theta*= 0.654,
z = 364.841
[jh_simplexP] Iteracio 16 : iout = 7, q = 2, B(p) = 1, theta*= 0.104,
z = 187.990
[jh_simplexP] Iteracio 17 : iout = 6, q = 9, B(p) = 5, theta*= 1.050,
z = 157.658
[jh_simplexP] Iteracio 18 : iout = 2, q = 13, B(p) = 22, theta*= 0.714,
z = 156.481
[jh_simplexP] Iteracio 19 : iout = -, q = -, B(p) = -, theta*= -,
z = 0.000
[jh_simplexP] Solucio basica factible trobada, iteracio 19
[jh_simplexP] Fase II
```

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[jh_simplexP] Iteracio 20 : iout = 6, q = 5, B(p) = 9, theta*= 1.706,
z = -77.097
[jh_simplexP] Iteracio 21 : iout = 5, q = 15, B(p) = 8, theta*=
179.166, z = -132.510
[jh_simplexP] Iteracio 22 : iout = 9, q = 16, B(p) = 4, theta*= 9.381,
z = -245.214
[jh_simplexP] Iteracio 23 : iout = 5, q = 1, B(p) = 15, theta*= 1.397,
z = -251.992
[jh_simplexP] Iteracio 24 : iout = 6, q = 8, B(p) = 5, theta*= 0.702,
z = -296.160
[jh_simplexP] Iteracio 25 : iout = 7, q = 17, B(p) = 2, theta*=
273.616, z = -323.140
[jh_simplexP] Iteracio 26 : iout = 1, q = 19, B(p) = 6, theta*=
147.251, z = -363.300
[jh_simplexP] Iteracio 27 : iout = 5, q = 2, B(p) = 1, theta*= 0.761,
z = -531.478
[jh_simplexP] Iteracio 28 : iout = 7, q = 5, B(p) = 17, theta*= 0.832,
z = -580.361
[jh_simplexP] Iteracio 29 : iout = 3, q = 7, B(p) = 11, theta*= 1.360,
z = -588.083
[jh_simplexP] Iteracio 30 : iout = 7, q = 17, B(p) = 5, theta*=
356.161, z = -630.546
[jh_simplexP] Iteracio 31 : iout = 5, q = 20, B(p) = 2, theta*=
104.747, z = -639.085
[jh_simplexP] Iteracio 32 : iout = 2, q = 5, B(p) = 13, theta*= 2.203,
z = -695.736
[jh_simplexP] Iteracio 33 : iout = 7, q = 1, B(p) = 17, theta*= 0.004,
z = -787.471
[jh_simplexP] Iteracio 34 : iout = -, q = -, B(p) = -, theta*= -,
z = -787.487
[jh_simplexP] Solucio optima trobada, iteracio 34, z = -787.486630
[jh_simplexP] Fi simplex primal

```

Solucio optima:

```

vb = 19  5  7 10 20  8  1 12 16  3
xb = 764.4    2.2    1.5    2.5 291.8    3.6    0.0    2.2 322.8    0.5
z = -787.4866
r =    0.0    31.4    0.0 110.0   -0.0 192.2   -0.0    0.0 176.1   -0.0    5.5
    0.0 114.3 186.0    0.4    0.0    0.0    0.5    0.0    0.0

```

Problema 2

```

[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 1, q = 1, B(p) = 21, theta*= 0.113,
z = 2231.000
[jh_simplexP] Iteracio 2 : iout = 8, q = 3, B(p) = 28, theta*= 0.030,
z = 2223.189
[jh_simplexP] Iteracio 3 : iout = 6, q = 4, B(p) = 26, theta*= 0.951,
z = 2208.520

```

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[jh_simplexP] Iteracio 4 : iout = 10, q = 2, B(p) = 30, theta*= 0.145,
z = 1766.561
[jh_simplexP] Iteracio 5 : iout = 5, q = 5, B(p) = 25, theta*= 2.494,
z = 1724.330
[jh_simplexP] Iteracio 6 : iout = 1, q = 6, B(p) = 1, theta*= 0.132,
z = 1441.831
[jh_simplexP] Iteracio 7 : iout = 1, q = 7, B(p) = 6, theta*= 0.108,
z = 1412.946
[jh_simplexP] Iteracio 8 : iout = 9, q = 8, B(p) = 29, theta*= 0.008,
z = 1393.373
[jh_simplexP] Iteracio 9 : iout = 1, q = 6, B(p) = 7, theta*= 0.666,
z = 1379.835
[jh_simplexP] Iteracio 10 : iout = 4, q = 10, B(p) = 24, theta*= 0.394,
z = 1314.303
[jh_simplexP] Iteracio 11 : iout = 8, q = 1, B(p) = 3, theta*= 2.818,
z = 894.204
[jh_simplexP] Iteracio 12 : iout = 10, q = 7, B(p) = 2, theta*= 1.790,
z = 888.772
[jh_simplexP] Iteracio 13 : iout = 8, q = 3, B(p) = 1, theta*= 1.718,
z = 411.338
[jh_simplexP] Iteracio 14 : iout = 2, q = 9, B(p) = 22, theta*= 0.237,
z = 392.790
[jh_simplexP] Iteracio 15 : iout = 3, q = 11, B(p) = 23, theta*= 0.050,
z = 313.545
[jh_simplexP] Iteracio 16 : iout = 2, q = 1, B(p) = 9, theta*= 0.049,
z = 282.348
[jh_simplexP] Iteracio 17 : iout = 6, q = 2, B(p) = 4, theta*= 1.079,
z = 278.716
[jh_simplexP] Iteracio 18 : iout = 7, q = 13, B(p) = 27, theta*= 1.030,
z = 136.476
[jh_simplexP] Iteracio 19 : iout = -, q = -, B(p) = -, theta*= -,
z = 0.000
[jh_simplexP] Solucio basica factible trobada, iteracio 19
[jh_simplexP] Fase II
[jh_simplexP] Iteracio 20 : iout = 10, q = 9, B(p) = 7, theta*= 2.023,
z = 417.139
[jh_simplexP] Iteracio 21 : iout = 1, q = 12, B(p) = 6, theta*= 2.362,
z = 267.479
[jh_simplexP] Iteracio 22 : iout = 8, q = 7, B(p) = 3, theta*= 0.057,
z = 135.921
[jh_simplexP] Iteracio 23 : iout = 5, q = 14, B(p) = 5, theta*= 0.100,
z = 119.775
[jh_simplexP] Iteracio 24 : iout = 8, q = 3, B(p) = 7, theta*= 0.520,
z = 108.443
[jh_simplexP] Iteracio 25 : iout = 9, q = 4, B(p) = 8, theta*= 0.316,
z = 82.939
[jh_simplexP] Iteracio 26 : iout = 9, q = 6, B(p) = 4, theta*= 0.180,
z = 56.912
[jh_simplexP] Iteracio 27 : iout = 9, q = 15, B(p) = 6, theta*=
105.586, z = 54.896
[jh_simplexP] Iteracio 28 : iout = 4, q = 16, B(p) = 10, theta*= 73.570,
z = 53.658
[jh_simplexP] Iteracio 29 : iout = 10, q = 4, B(p) = 9, theta*= 1.514,
z = 18.470

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[jh_simplexP] Iteracio 30 : iout = 10, q = 6, B(p) = 4, theta*= 0.622,
z = -23.129
[jh_simplexP] Iteracio 31 : iout = 9, q = 10, B(p) = 15, theta*= 0.926,
z = -32.778
[jh_simplexP] Iteracio 32 : iout = 9, q = 17, B(p) = 10, theta*=
150.172, z = -42.298
[jh_simplexP] Iteracio 33 : iout = 8, q = 19, B(p) = 3, theta*=
444.748, z = -93.057
[jh_simplexP] Iteracio 34 : iout = -, q = -, B(p) = -, theta*= -,
z = -250.134
[jh_simplexP] Solucio optima trobada, iteracio 34, z = -250.133658
[jh_simplexP] Fi simplex primal

```

Solucio optima:

```

vb = 12  1 11 16 14  2 13 19 17  6
xb =   1.4  2.4  0.3 294.0  4.3  0.1  3.0 444.7 520.0  2.7
z = -250.1337
r =   0.0 -0.0 90.3 16.0 32.7  0.0 189.2 91.8 48.4 57.6  0.0
    0.0 -0.0 -0.0  0.2  0.0  0.0  0.5  0.0  0.4

```

Problema 3

```

[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 8, q = 1, B(p) = 28, theta*= 0.362,
z = 1121.000
[jh_simplexP] Iteracio 2 : iout = 10, q = 2, B(p) = 30, theta*= 0.072,
z = 1107.950
[jh_simplexP] Iteracio 3 : iout = 8, q = 3, B(p) = 1, theta*= 0.162,
z = 1103.248
[jh_simplexP] Iteracio 4 : iout = 10, q = 4, B(p) = 2, theta*= 0.056,
z = 1052.098
[jh_simplexP] Iteracio 5 : iout = 6, q = 5, B(p) = 26, theta*= 0.487,
z = 1031.453
[jh_simplexP] Iteracio 6 : iout = 10, q = 1, B(p) = 4, theta*= 0.008,
z = 951.743
[jh_simplexP] Iteracio 7 : iout = 10, q = 2, B(p) = 1, theta*= 0.006,
z = 950.516
[jh_simplexP] Iteracio 8 : iout = 10, q = 8, B(p) = 2, theta*= 0.006,
z = 950.032
[jh_simplexP] Iteracio 9 : iout = 10, q = 16, B(p) = 8, theta*= 0.809,
z = 949.273
[jh_simplexP] Iteracio 10 : iout = 7, q = 17, B(p) = 27, theta*=
293.523, z = 948.020
[jh_simplexP] Iteracio 11 : iout = 10, q = 12, B(p) = 16, theta*= 0.112,
z = 654.497
[jh_simplexP] Iteracio 12 : iout = 8, q = 18, B(p) = 3, theta*= 11.785,
z = 638.813
[jh_simplexP] Iteracio 13 : iout = 9, q = 19, B(p) = 29, theta*= 48.817,
z = 621.054

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[jh_simplexP] Iteracio 14 : iout = 6, q = 26, B(p) = 5, theta*= 2.444,
z = 572.237
[jh_simplexP] Iteracio 15 : iout = -, q = -, B(p) = -, theta*= -,
z = 471.778
[jh_simplexP] Problema infactible.
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Problema 4

```
[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 8, q = 1, B(p) = 32, theta*= 8.155,
z = 7315.000
[jh_simplexP] Iteracio 2 : iout = 7, q = 2, B(p) = 31, theta*= 2.340,
z = 3767.732
[jh_simplexP] Iteracio 3 : iout = 9, q = 3, B(p) = 33, theta*= 6.480,
z = 2774.501
[jh_simplexP] Iteracio 4 : iout = 2, q = 5, B(p) = 26, theta*= 0.965,
z = 2305.679
[jh_simplexP] Iteracio 5 : iout = 6, q = 4, B(p) = 30, theta*= 0.222,
z = 1933.786
[jh_simplexP] Iteracio 6 : iout = 9, q = 8, B(p) = 3, theta*= 0.099,
z = 392.172
[jh_simplexP] Iteracio 7 : iout = 6, q = 6, B(p) = 4, theta*= 0.225,
z = 372.775
[jh_simplexP] Iteracio 8 : iout = 6, q = 7, B(p) = 6, theta*= 1.002,
z = 368.020
[jh_simplexP] Iteracio 9 : iout = 3, q = 9, B(p) = 27, theta*= 0.214,
z = 313.193
[jh_simplexP] Iteracio 10 : iout = 1, q = 10, B(p) = 25, theta*= 0.312,
z = 276.343
[jh_simplexP] Iteracio 11 : iout = 5, q = 3, B(p) = 29, theta*= 1.031,
z = 156.435
[jh_simplexP] Iteracio 12 : iout = 5, q = 6, B(p) = 3, theta*= 0.486,
z = 110.614
[jh_simplexP] Iteracio 13 : iout = 10, q = 11, B(p) = 34, theta*= 0.251,
z = 61.507
[jh_simplexP] Iteracio 14 : iout = 5, q = 3, B(p) = 6, theta*= 0.593,
z = 27.792
[jh_simplexP] Iteracio 15 : iout = 4, q = 12, B(p) = 28, theta*= 0.423,
z = 25.700
[jh_simplexP] Iteracio 16 : iout = -, q = -, B(p) = -, theta*= -,
z = 0.000
[jh_simplexP] Solucio basica factible trobada, iteracio 16
[jh_simplexP] Fase II
[jh_simplexP] Iteracio 17 : iout = 5, q = 6, B(p) = 3, theta*= 0.687,
z = -774.572
[jh_simplexP] Iteracio 18 : iout = 3, q = 13, B(p) = 9, theta*= 0.924,
z = -894.881
[jh_simplexP] Iteracio 19 : iout = 1, q = 14, B(p) = 10, theta*= 0.423,
z = -982.164
[jh_simplexP] Iteracio 20 : iout = 4, q = 15, B(p) = 12, theta*= 2.057,
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z = -1072.642
[jh_simplexP] Iteracio 21 : iout = 4, q = 16, B(p) = 15, theta*= 1.275,
z = -1073.653
[jh_simplexP] Iteracio 22 : iout = 6, q = 18, B(p) = 7, theta*= 18.095,
z = -1074.433
[jh_simplexP] Iteracio 23 : iout = 5, q = 9, B(p) = 6, theta*= 0.307,
z = -1088.044
[jh_simplexP] Iteracio 24 : iout = 7, q = 15, B(p) = 2, theta*= 45.483,
z = -1103.173
[jh_simplexP] Iteracio 25 : iout = 5, q = 6, B(p) = 9, theta*= 0.122,
z = -1122.540
[jh_simplexP] Iteracio 26 : iout = 5, q = 20, B(p) = 6, theta*= 5.499,
z = -1123.208
[jh_simplexP] Iteracio 27 : iout = 7, q = 2, B(p) = 15, theta*= 1.014,
z = -1128.115
[jh_simplexP] Iteracio 28 : iout = 9, q = 17, B(p) = 8, theta*= 91.735,
z = -1132.582
[jh_simplexP] Iteracio 29 : iout = 7, q = 15, B(p) = 2, theta*=
200.097, z = -1183.095
[jh_simplexP] Iteracio 30 : iout = 10, q = 7, B(p) = 11, theta*= 3.975,
z = -1229.467
[jh_simplexP] Iteracio 31 : iout = 8, q = 19, B(p) = 1, theta*=
198.423, z = -1259.729
[jh_simplexP] Iteracio 32 : iout = 10, q = 11, B(p) = 7, theta*= 0.361,
z = -1335.323
[jh_simplexP] Iteracio 33 : iout = 10, q = 21, B(p) = 11, theta*= 30.122,
z = -1337.273
[jh_simplexP] Iteracio 34 : iout = 3, q = 22, B(p) = 13, theta*=
1934.565, z = -1383.652
[jh_simplexP] Iteracio 35 : iout = 2, q = 7, B(p) = 5, theta*= 6.267,
z = -2468.023
[jh_simplexP] Iteracio 36 : iout = 2, q = 24, B(p) = 7, theta*=
564.000, z = -2594.600
[jh_simplexP] Iteracio 37 : iout = 3, q = 13, B(p) = 22, theta*= 85.241,
z = -4625.000
[jh_simplexP] Iteracio 38 : iout = 1, q = 23, B(p) = 14, theta*=
631.500, z = -5477.414
[jh_simplexP] Iteracio 39 : problema no acotat detectat.
[jh_simplexP] Problema no acotat.

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Conjunt 37

Problema 1

```

[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 7, q = 1, B(p) = 27, theta*= 0.072,
z = 2278.000
[jh_simplexP] Iteracio 2 : iout = 6, q = 2, B(p) = 26, theta*= 3.391,
z = 2261.402
[jh_simplexP] Iteracio 3 : iout = 3, q = 4, B(p) = 23, theta*= 1.038,

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z = 1709.298
[jh_simplexP] Iteracio 4 : iout = 2, q = 3, B(p) = 22, theta*= 0.038,
z = 1382.148
[jh_simplexP] Iteracio 5 : iout = 2, q = 5, B(p) = 3, theta*= 0.035,
z = 1366.560
[jh_simplexP] Iteracio 6 : iout = 10, q = 6, B(p) = 30, theta*= 0.166,
z = 1361.862
[jh_simplexP] Iteracio 7 : iout = 7, q = 3, B(p) = 1, theta*= 0.183,
z = 1325.968
[jh_simplexP] Iteracio 8 : iout = 1, q = 7, B(p) = 21, theta*= 0.663,
z = 1291.896
[jh_simplexP] Iteracio 9 : iout = 10, q = 1, B(p) = 6, theta*= 0.012,
z = 1106.070
[jh_simplexP] Iteracio 10 : iout = 9, q = 9, B(p) = 29, theta*= 1.593,
z = 1099.446
[jh_simplexP] Iteracio 11 : iout = 5, q = 10, B(p) = 25, theta*= 0.773,
z = 584.210
[jh_simplexP] Iteracio 12 : iout = 8, q = 11, B(p) = 28, theta*= 1.221,
z = 277.690
[jh_simplexP] Iteracio 13 : iout = 10, q = 6, B(p) = 1, theta*= 0.127,
z = 65.586
[jh_simplexP] Iteracio 14 : iout = 10, q = 8, B(p) = 6, theta*= 0.188,
z = 46.863
[jh_simplexP] Iteracio 15 : iout = 7, q = 12, B(p) = 3, theta*= 0.864,
z = 32.430
[jh_simplexP] Iteracio 16 : iout = 4, q = 13, B(p) = 24, theta*= 0.640,
z = 25.819
[jh_simplexP] Iteracio 17 : iout = -, q = -, B(p) = -, theta*= -,
z = 0.000
[jh_simplexP] Solucio basica factible trobada, iteracio 17
[jh_simplexP] Fase II
[jh_simplexP] Iteracio 18 : iout = 4, q = 14, B(p) = 13, theta*= 0.065,
z = 183.243
[jh_simplexP] Iteracio 19 : iout = 2, q = 6, B(p) = 5, theta*= 0.572,
z = 163.725
[jh_simplexP] Iteracio 20 : iout = 1, q = 15, B(p) = 7, theta*=
374.780, z = 145.879
[jh_simplexP] Iteracio 21 : iout = 9, q = 16, B(p) = 9, theta*=
111.238, z = 51.560
[jh_simplexP] Iteracio 22 : iout = 7, q = 5, B(p) = 12, theta*= 0.618,
z = 7.871
[jh_simplexP] Iteracio 23 : iout = 5, q = 17, B(p) = 10, theta*=
100.232, z = -5.342
[jh_simplexP] Iteracio 24 : iout = 10, q = 7, B(p) = 8, theta*= 0.628,
z = -30.623
[jh_simplexP] Iteracio 25 : iout = 7, q = 12, B(p) = 5, theta*= 0.181,
z = -46.593
[jh_simplexP] Iteracio 26 : iout = 10, q = 9, B(p) = 7, theta*= 0.214,
z = -57.585
[jh_simplexP] Iteracio 27 : iout = 1, q = 19, B(p) = 15, theta*=
529.954, z = -60.506
[jh_simplexP] Iteracio 28 : iout = 3, q = 8, B(p) = 4, theta*= 0.465,
z = -172.346
[jh_simplexP] Iteracio 29 : iout = 3, q = 13, B(p) = 8, theta*= 1.365,

```

```

z = -193.197
[jh_simplexP] Iteracio 30 : iout = -, q = -, B(p) = -, theta*= -,
z = -197.476
[jh_simplexP] Solucio optima trobada, iteracio 30, z = -197.476081
[jh_simplexP] Fi simplex primal

```

Solucio optima:

```

vb = 19 6 13 14 17 2 12 11 16 9
xb = 651.2 3.4 1.4 0.4 433.8 6.6 0.8 2.9 1038.4 0.2
z = -197.4761
r = 152.8 0.0 43.4 25.9 52.8 0.0 2.4 9.2 -0.0 166.9 0.0
0.0 0.0 0.0 0.2 0.0 0.0 0.2 0.0 0.7

```

Problema 2

```

[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 2, q = 1, B(p) = 22, theta*= 0.377,
z = 2625.000
[jh_simplexP] Iteracio 2 : iout = 2, q = 2, B(p) = 1, theta*= 0.684,
z = 2498.014
[jh_simplexP] Iteracio 3 : iout = 6, q = 4, B(p) = 26, theta*= 0.166,
z = 2334.211
[jh_simplexP] Iteracio 4 : iout = 3, q = 1, B(p) = 23, theta*= 0.175,
z = 2143.096
[jh_simplexP] Iteracio 5 : iout = 1, q = 3, B(p) = 21, theta*= 0.384,
z = 2040.842
[jh_simplexP] Iteracio 6 : iout = 7, q = 5, B(p) = 27, theta*= 0.864,
z = 1945.333
[jh_simplexP] Iteracio 7 : iout = 4, q = 6, B(p) = 24, theta*= 1.436,
z = 1801.254
[jh_simplexP] Iteracio 8 : iout = 9, q = 7, B(p) = 29, theta*= 1.060,
z = 1641.509
[jh_simplexP] Iteracio 9 : iout = 8, q = 9, B(p) = 28, theta*= 0.162,
z = 855.444
[jh_simplexP] Iteracio 10 : iout = 5, q = 8, B(p) = 25, theta*= 1.387,
z = 717.123
[jh_simplexP] Iteracio 11 : iout = 6, q = 10, B(p) = 4, theta*= 0.473,
z = 184.750
[jh_simplexP] Iteracio 12 : iout = 5, q = 11, B(p) = 8, theta*= 1.371,
z = 156.310
[jh_simplexP] Iteracio 13 : iout = 1, q = 12, B(p) = 3, theta*= 0.069,
z = 49.013
[jh_simplexP] Iteracio 14 : iout = 4, q = 4, B(p) = 6, theta*= 0.256,
z = 39.014
[jh_simplexP] Iteracio 15 : iout = 9, q = 3, B(p) = 7, theta*= 0.230,
z = 18.269
[jh_simplexP] Iteracio 16 : iout = 10, q = 8, B(p) = 30, theta*= 1.620,
z = 13.140
[jh_simplexP] Iteracio 17 : iout = -, q = -, B(p) = -, theta*= -,

```

```

z = 0.000
[jh_simplexP] Solucio basica factible trobada, iteracio 17
[jh_simplexP] Fase II
[jh_simplexP] Iteracio 18 : iout = 6, q = 6, B(p) = 10, theta*= 0.206,
z = -246.297
[jh_simplexP] Iteracio 19 : iout = 9, q = 7, B(p) = 3, theta*= 0.339,
z = -519.979
[jh_simplexP] Iteracio 20 : iout = 6, q = 14, B(p) = 6, theta*= 0.357,
z = -537.351
[jh_simplexP] Iteracio 21 : iout = 7, q = 3, B(p) = 5, theta*= 0.181,
z = -557.947
[jh_simplexP] Iteracio 22 : iout = 6, q = 15, B(p) = 14, theta*= 47.146,
z = -562.608
[jh_simplexP] Iteracio 23 : iout = 6, q = 20, B(p) = 15, theta*= 14.253,
z = -580.288
[jh_simplexP] Iteracio 24 : iout = -, q = -, B(p) = -, theta*= -,
z = -586.610
[jh_simplexP] Solucio optima trobada, iteracio 24, z = -586.610343
[jh_simplexP] Fi simplex primal

```

Solucio optima:

```

vb = 12  2  1  4 11 20  3  9  7  8
xb =   3.1   0.7   1.8   1.5   1.2  14.3   0.3   1.5   0.0   3.4
z = -586.6103
r =   0.0   0.0  -0.0  -0.0  40.5 304.0  -0.0   0.0  -0.0 248.2   0.0
0.0  83.2  44.7   0.1   1.5   1.3   0.8   1.3   0.0

```

Problema 3

```

[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 8, q = 1, B(p) = 28, theta*= 0.586,
z = 1801.000
[jh_simplexP] Iteracio 2 : iout = 8, q = 3, B(p) = 1, theta*= 0.667,
z = 1618.103
[jh_simplexP] Iteracio 3 : iout = 9, q = 5, B(p) = 29, theta*= 0.363,
z = 1560.333
[jh_simplexP] Iteracio 4 : iout = 5, q = 10, B(p) = 25, theta*= 0.556,
z = 1501.739
[jh_simplexP] Iteracio 5 : iout = 5, q = 15, B(p) = 10, theta*= 58.814,
z = 1472.841
[jh_simplexP] Iteracio 6 : iout = 7, q = 17, B(p) = 27, theta*= 82.953,
z = 1442.925
[jh_simplexP] Iteracio 7 : iout = 9, q = 4, B(p) = 5, theta*= 0.660,
z = 1359.973
[jh_simplexP] Iteracio 8 : iout = 8, q = 10, B(p) = 3, theta*= 0.422,
z = 1343.945
[jh_simplexP] Iteracio 9 : iout = 5, q = 19, B(p) = 15, theta*= 19.867,
z = 1342.596
[jh_simplexP] Iteracio 10 : iout = 9, q = 3, B(p) = 4, theta*= 0.080,

```

```

z = 1340.436
[jh_simplexP] Iteracio 11 : iout = 10, q = 20, B(p) = 30, theta*= 56.709,
z = 1340.237
[jh_simplexP] Iteracio 12 : iout = 9, q = 4, B(p) = 3, theta*= 0.085,
z = 1283.528
[jh_simplexP] Iteracio 13 : iout = -, q = -, B(p) = -, theta*= -,
z = 1282.453
[jh_simplexP] Problema infactible.

```

Problema 4

```

[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 8, q = 1, B(p) = 32, theta*= 8.866,
z = 6828.000
[jh_simplexP] Iteracio 2 : iout = 5, q = 2, B(p) = 29, theta*= 4.076,
z = 2483.670
[jh_simplexP] Iteracio 3 : iout = 5, q = 3, B(p) = 2, theta*= 1.467,
z = 2397.856
[jh_simplexP] Iteracio 4 : iout = 2, q = 4, B(p) = 26, theta*= 2.368,
z = 2050.292
[jh_simplexP] Iteracio 5 : iout = 4, q = 5, B(p) = 28, theta*= 3.489,
z = 1347.373
[jh_simplexP] Iteracio 6 : iout = 2, q = 2, B(p) = 4, theta*= 0.034,
z = 875.187
[jh_simplexP] Iteracio 7 : iout = 2, q = 7, B(p) = 2, theta*= 0.102,
z = 813.890
[jh_simplexP] Iteracio 8 : iout = 2, q = 8, B(p) = 7, theta*= 0.162,
z = 813.641
[jh_simplexP] Iteracio 9 : iout = 6, q = 9, B(p) = 30, theta*= 0.954,
z = 783.651
[jh_simplexP] Iteracio 10 : iout = 10, q = 4, B(p) = 34, theta*= 0.052,
z = 506.720
[jh_simplexP] Iteracio 11 : iout = 10, q = 6, B(p) = 4, theta*= 0.078,
z = 489.532
[jh_simplexP] Iteracio 12 : iout = 10, q = 7, B(p) = 6, theta*= 0.165,
z = 485.329
[jh_simplexP] Iteracio 13 : iout = 2, q = 10, B(p) = 8, theta*= 0.251,
z = 483.995
[jh_simplexP] Iteracio 14 : iout = 2, q = 11, B(p) = 10, theta*= 0.158,
z = 481.791
[jh_simplexP] Iteracio 15 : iout = 10, q = 4, B(p) = 7, theta*= 1.330,
z = 451.599
[jh_simplexP] Iteracio 16 : iout = 1, q = 6, B(p) = 25, theta*= 1.977,
z = 412.496
[jh_simplexP] Iteracio 17 : iout = 7, q = 12, B(p) = 31, theta*= 0.711,
z = 248.833
[jh_simplexP] Iteracio 18 : iout = 5, q = 10, B(p) = 3, theta*= 0.782,
z = 204.689
[jh_simplexP] Iteracio 19 : iout = 9, q = 13, B(p) = 33, theta*= 0.592,
z = 187.369

```

```

[jh_simplexP] Iteracio 20 : iout = 3, q = 2, B(p) = 27, theta*= 0.071,
z = 7.419
[jh_simplexP] Iteracio 21 : iout = -, q = -, B(p) = -, theta*= -,
z = 0.000
[jh_simplexP] Solucio basica factible trobada, iteracio 21
[jh_simplexP] Fase II
[jh_simplexP] Iteracio 22 : iout = 3, q = 7, B(p) = 2, theta*= 0.102,
z = -788.970
[jh_simplexP] Iteracio 23 : iout = 8, q = 8, B(p) = 1, theta*= 0.999,
z = -791.546
[jh_simplexP] Iteracio 24 : iout = 7, q = 14, B(p) = 12, theta*= 0.473,
z = -876.151
[jh_simplexP] Iteracio 25 : iout = 10, q = 2, B(p) = 4, theta*= 0.153,
z = -908.193
[jh_simplexP] Iteracio 26 : iout = 10, q = 17, B(p) = 2, theta*=
110.029, z = -915.049
[jh_simplexP] Iteracio 27 : iout = 4, q = 12, B(p) = 5, theta*= 0.045,
z = -934.298
[jh_simplexP] Iteracio 28 : iout = 2, q = 18, B(p) = 11, theta*= 83.214,
z = -941.734
[jh_simplexP] Iteracio 29 : iout = 7, q = 4, B(p) = 14, theta*= 0.084,
z = -975.353
[jh_simplexP] Iteracio 30 : iout = 3, q = 15, B(p) = 7, theta*= 7.563,
z = -979.763
[jh_simplexP] Iteracio 31 : iout = 7, q = 19, B(p) = 4, theta*= 18.090,
z = -981.786
[jh_simplexP] Iteracio 32 : iout = 4, q = 11, B(p) = 12, theta*= 0.546,
z = -993.564
[jh_simplexP] Iteracio 33 : iout = 4, q = 16, B(p) = 11, theta*= 75.522,
z = -1017.486
[jh_simplexP] Iteracio 34 : iout = 10, q = 4, B(p) = 17, theta*= 0.182,
z = -1025.912
[jh_simplexP] Iteracio 35 : iout = 9, q = 20, B(p) = 13, theta*= 75.212,
z = -1025.990
[jh_simplexP] Iteracio 36 : iout = 10, q = 2, B(p) = 4, theta*= 1.770,
z = -1099.114
[jh_simplexP] Iteracio 37 : iout = 8, q = 17, B(p) = 8, theta*=
234.920, z = -1136.402
[jh_simplexP] Iteracio 38 : iout = 10, q = 21, B(p) = 2, theta*= 46.640,
z = -1290.780
[jh_simplexP] Iteracio 39 : iout = 6, q = 11, B(p) = 9, theta*= 3.879,
z = -1341.322
[jh_simplexP] Iteracio 40 : iout = 6, q = 22, B(p) = 11, theta*=
1170.705, z = -1410.919
[jh_simplexP] Iteracio 41 : iout = 5, q = 2, B(p) = 10, theta*= 16.187,
z = -2293.982
[jh_simplexP] Iteracio 42 : iout = 9, q = 13, B(p) = 20, theta*= 3.701,
z = -2384.052
[jh_simplexP] Iteracio 43 : iout = 5, q = 23, B(p) = 2, theta*=
167.687, z = -2725.429
[jh_simplexP] Iteracio 44 : iout = 7, q = 20, B(p) = 19, theta*=
725.863, z = -2962.109
[jh_simplexP] Iteracio 45 : iout = 1, q = 2, B(p) = 6, theta*= 1.983,
z = -4267.537

```

```
[jh_simplexP] Iteracio 46 : iout = 1, q = 24, B(p) = 2, theta*=  
132.077, z = -4486.508  
[jh_simplexP] Iteracio 47 : iout = 9, q = 4, B(p) = 13, theta*=  
799.000, z = -5777.385  
[jh_simplexP] Iteracio 48 : problema no acotat detectat.  
[jh_simplexP] Problema no acotat.
```